

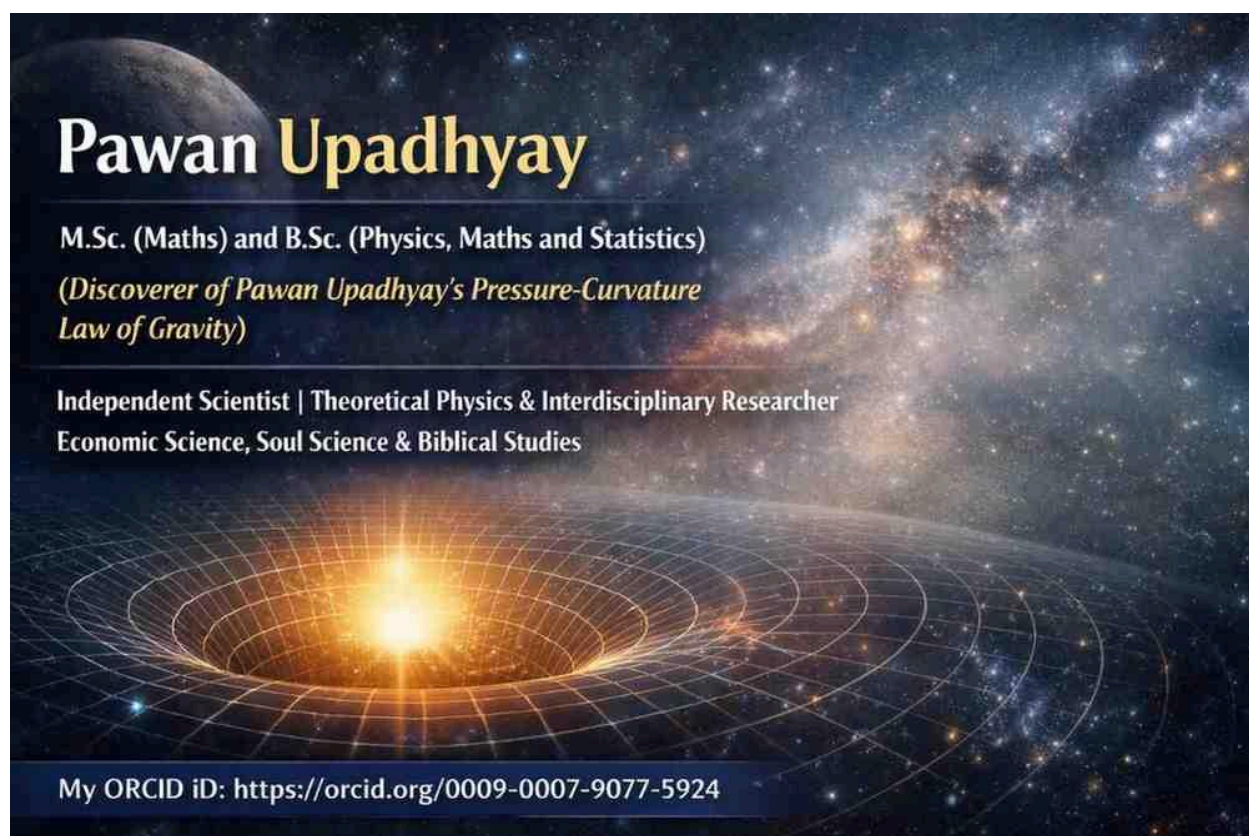
Circular Motion and Kepler's Laws in the PPC Law of Gravity

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Abstract

This paper develops a pressure-based interpretation of orbital motion within Pawan Upadhyay's Pressure Curvature (PPC) Law of Gravity. Starting from the PPC force-density relation, we derive the conditions under which circular motion and Kepler's laws emerge. We show that if the gravitational pressure field yields an inverse-square pressure gradient, then the standard

Keplerian results follow naturally. This establishes consistency of the PPC framework with classical celestial mechanics in the weak-field regime.

1. Introduction

In classical mechanics, orbital motion is explained by Newton's law of gravitation, while in relativistic physics it is described by spacetime curvature. The PPC law introduces an intermediate mechanism: gravitational pressure.

The central idea is that:

Mass Density $\rightarrow$ Energy Density $\rightarrow$ Pressure $\rightarrow$ Pressure Gradient $\rightarrow$ Motion

This paper focuses on how circular motion and Kepler's laws arise from this pressure-based formulation.

2. Fundamental PPC Equation

The PPC force-density relation is:

$$\rho \, dv/dt = -\nabla P_g$$

For circular motion, acceleration is centripetal:

$$a = v^2 / r$$

Thus, the PPC equation becomes:

$$\rho \, v^2 / r = -\nabla P_g$$

This equation states that the centripetal force density required for circular motion is provided by the gradient of gravitational pressure.

3. Radial Pressure Field and Inverse-Square Law

To recover physically observed orbits, the pressure field must satisfy:

- $(1/\rho) \, dP_g/dr = GM / r^2$

Integrating gives:

$$P_g(r) = \rho \Phi(r)$$

where $\Phi(r) = -GM/r$ is the gravitational potential.

Thus, gravitational pressure is directly proportional to potential in the weak-field limit.

4. Circular Orbital Velocity

Substituting into the PPC equation:

$$\rho v^2 / r = \rho GM / r^2$$

Cancelling ρ :

$$v^2 = GM / r$$

This is the standard expression for circular orbital velocity.

5. Kepler's Third Law

For circular motion:

$$v = 2\pi r / T$$

Substituting into $v^2 = GM/r$:

$$(2\pi r/T)^2 = GM/r$$

Solving for T:

$$T^2 = (4\pi^2 / GM) r^3$$

This is Kepler's Third Law.

6. Kepler's Second Law (Equal Areas)

Since the pressure gradient force is radial:

$$\mathbf{F} = -\nabla P_g \text{ (}\hat{r}\text{ direction)}$$

There is no tangential component of force. Therefore, angular momentum is conserved:

$$\mathbf{L} = m \mathbf{r} \times \mathbf{v} = \text{constant}$$

This implies that the areal velocity is constant:

$$dA/dt = \text{constant}$$

Thus, the radius vector sweeps equal areas in equal times.

7. Kepler's First Law (Elliptical Orbits)

A central force proportional to $1/r^2$ yields conic-section trajectories. Since PPC reproduces:

$$a(r) = GM / r^2$$

it follows that bound orbits are ellipses with the central mass at one focus.

8. Physical Interpretation in PPC

- Mass–energy generates pressure P_g
- Pressure gradients produce force
- Radial pressure distribution yields inverse-square behavior
- This leads to stable orbital motion

Thus, Keplerian dynamics emerge as a natural consequence of pressure-driven motion.

9. Weak-Field Consistency

In weak gravitational fields:

- Spacetime curvature is small
- PPC reduces to classical mechanics
- Pressure gradients act as effective gravitational forces

This ensures compatibility with observed planetary motion.

10. Conclusion

The PPC Law of Gravity successfully reproduces circular motion and all three of Kepler's laws when the gravitational pressure field generates an inverse-square gradient. This demonstrates that orbital dynamics can be understood as a consequence of pressure-driven forces, providing an alternative physical interpretation of gravity.

Keywords

PPC Gravity, Gravitational Pressure, Kepler's Laws, Circular Motion, Orbital Mechanics, Pressure Gradient, Inverse-Square Law